EEG-to-Video Retrieval Using Temporal Audiovisual Representations

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Evaluation	Top-1	Top-5	MRR
Final v7 DEV: 408 queries, 102 candidates	72.55%	93.87%	0.8196
Training-video validation: independent ranking	29.88%	54.11%	0.4173
Training-video validation: CSLS and assignment	29.15%	58.25%	0.4257

Table 1: Retrieval performance under distinct evaluation protocols. DEV metrics pool 408 queries from subjects 01–04, with 102 candidates per query. Training-video validation reports unweighted means of subject-level metrics across all six subjects, with 124–149 candidates per subject, fewer training videos, and fewer internal ensemble seeds. The absolute scores are therefore not directly comparable with DEV. Both DEV and training-video validation were repeatedly used during development and are not untouched evaluation sets.

1. Introduction

This paper investigates electroencephalography (EEG)-to-video retrieval in the *Mind Reading Using Machine Learning and EEG* challenge at the SchubKI hackathon. We describe the development of the final system, including the design decisions, negative results, and empirical observations that informed model selection.

The challenge is formulated as a constrained retrieval task: given an EEG recording, the goal is to identify the video that elicited it from a fixed set of audiovisual candidates. The task does not involve reconstructing arbitrary thoughts or generating unrestricted video content. The final system combines temporal audiovisual dual encoders with complementary audio and visual model scores. The fused scores undergo cross-domain similarity local scaling (CSLS), followed by one-to-one assignment. Our sequential experiments suggested that temporal representations and the training objective were important to retrieval performance; however, they do not isolate the causal contribution of each design choice.

Related work has also investigated candidate-score fusion for EEG- and MEG-to-image retrieval (8).

1.1 Task, Data, and Evaluation

The dataset comprises 1,000 training videos and 102 candidate test videos, each with a duration of three seconds. Training EEG recordings are available for all six subjects, although not every subject has a recording for every training video. Test EEG–video correspondences are available for subjects 01–04, whose 408 test recordings form the development set (DEV). Correspondences for subjects 05–06 are withheld for blind evaluation. Each subject has 102 test recordings, with one recording for each candidate video. Each EEG recording contains 17 channels and 1,500 samples acquired at a sampling rate of 500 Hz. For every recording, the submitted output is a complete ranking of candidate IDs 1001–1102, preserving the original order of the recordings.

Because training recordings are also available for subjects 05–06, the final evaluation concerns previously unseen stimuli for known subjects, rather than zero-shot generalization to entirely unseen subjects. In early leave-one-subject-out (LOSO) analyses, a subject was held out for configuration or fusion selection, while its EEG mapping was still fitted using that subject's own training recordings. These analyses therefore do not evaluate a model trained without access to the held-out subject.

We report Top-1 accuracy, Top-5 accuracy, and mean reciprocal rank (MRR). Top-k accuracy is the proportion of queries whose true candidate occurs among the first k ranks. Given the rank r_i of the true candidate for query i, MRR is defined as

$$\mathrm{MRR} = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{r_i},$$

where N is the number of queries in the evaluated set. For training-video validation, each metric is calculated separately for each subject and then averaged with equal subject weights. For uniformly random rankings among 102 candidates, expected Top-1 and Top-5 accuracies are $1/102 \approx 0.98\%$ and $5/102 \approx 4.90\%$, respectively. These chance levels apply to DEV and blind evaluation; training-video validation has different candidate counts. On DEV, one additional correct Top-1 prediction changes aggregate accuracy by approximately 0.245 percentage points. This numerical resolution does not quantify statistical uncertainty.

1.2 Hardware and Software

Model development and training took place during the hackathon on 17–18 October 2026. The primary computing resource was a bwUniCluster 3.0 node at the Karlsruhe Institute of Technology (KIT), equipped with four NVIDIA A100 GPUs with 80 GB of memory each. Additional model training used a local NVIDIA GeForce RTX 5070 Ti GPU. For collaborative work a private JupyterHub instance connected to a Python 3.11 kernel on the cluster node, was used.

The recorded release-verification environment was a separate Windows system running Python 3.11.9, NumPy 2.2.6, and PyTorch 2.14.1+cu130 on an NVIDIA GeForce RTX 5070 Ti. These versions describe the documented verification environment; they do not establish the exact software configuration of the earlier cluster runs. The release specifies its model dependencies in `requirements.txt` and pins the public encoder revisions in its configuration files..

1.3 Disclosure and Scope

For the final submission, challenge-specific training and fine-tuning used only official training recordings. Public pretrained encoders and supplied pretrained checkpoints were also used. Development labels informed evaluation, model selection, and other research decisions. Test EEG–video correspondences were not used as supervised training targets or label-dependent retrieval templates in the submitted system. Earlier exploratory methods that used labelled test responses were excluded from the final pipeline.

The internal ensemble normalises each subject's training and test sets separately using label-free statistics computed from each set. The supplied audio and visual models instead apply normalisation parameters fitted on training recordings. The final one-to-one assignment jointly processes the 102 test queries within each subject. Internal test-set normalisation and joint assignment make the final inference procedure transductive; encoder weights are not fitted on test recordings.

According to the team's account recorded in the project report, a jury clarification on 18 October 2026 permitted reproducible, label-free CSLS and one-to-one matching. The original clarification screenshots are not included in the release. External datasets and the provenance of supplied models are described in Section 8.

2. Approaches

2.1 EEG Data Analysis

We conducted exploratory analyses of the EEG recordings to examine their time-frequency structure and identify frequency ranges with stimulus-specific activity shared across subjects. These analyses informed subsequent modelling choices but did not directly estimate the performance of the final retrieval system.

2.1.1 Time-Frequency Analysis

We examined how spectral power varied over each three-second recording and compared normalisation schemes that emphasised different components of the response.

Method. We estimated spectral power in sliding windows of 0.5 s, advanced in steps of 50 ms. Within each window, we used discrete prolate spheroidal sequence (DPSS) tapers for spectral estimation and averaged the resulting power estimates across the 17 channels. The sequence of windowed estimates formed a time-frequency representation for each recording. The 50 ms step specifies the sampling of this representation in time; the 0.5 s window limits the temporal localisation of changes in power. Figure 1 illustrates individual recordings and a cross-subject summary for training video 4.

Normalisation and Observations. We compared three representations:

Absolute power. No reference normalisation was applied. Low-frequency power dominated the displayed scale, making relative temporal changes less apparent.

Relative to the mean spectrum. For each subject and frequency, power was divided by the mean power at that frequency across the subject's training recordings. The exploratory plots suggested an increase in relative power near the beginning of the videos in most subjects.

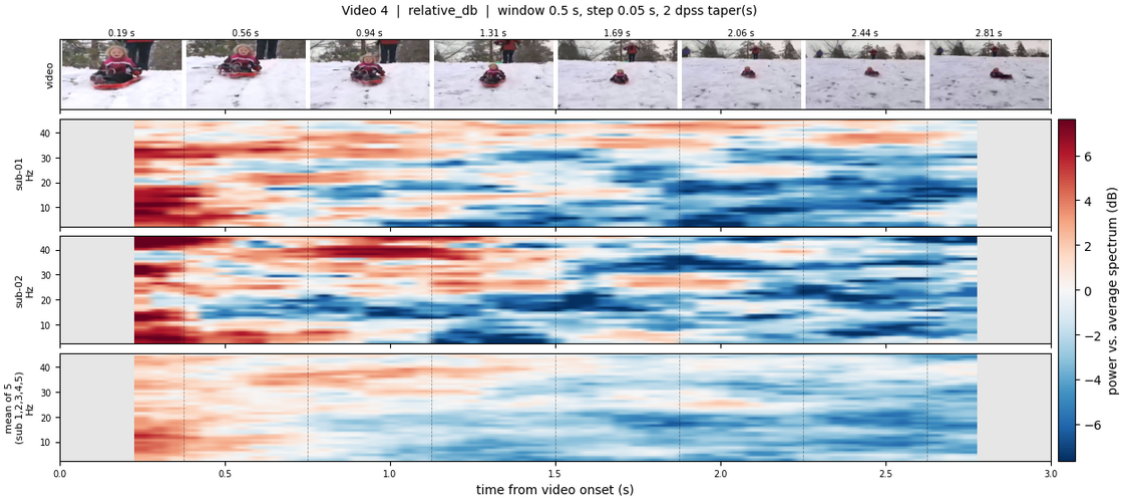
Relative to the mean time-frequency response. Power was divided by a subject-specific reference at each time and frequency, obtained by averaging across recordings. This reference accounts for temporal structure shared across stimuli, including a common onset response.

Under the time-frequency reference normalisation, the common onset pattern became less pronounced. The remaining patterns appeared more variable across subjects. Apparent alignment with events in individual videos was a qualitative observation rather than a quantified test of audiovisual tracking.

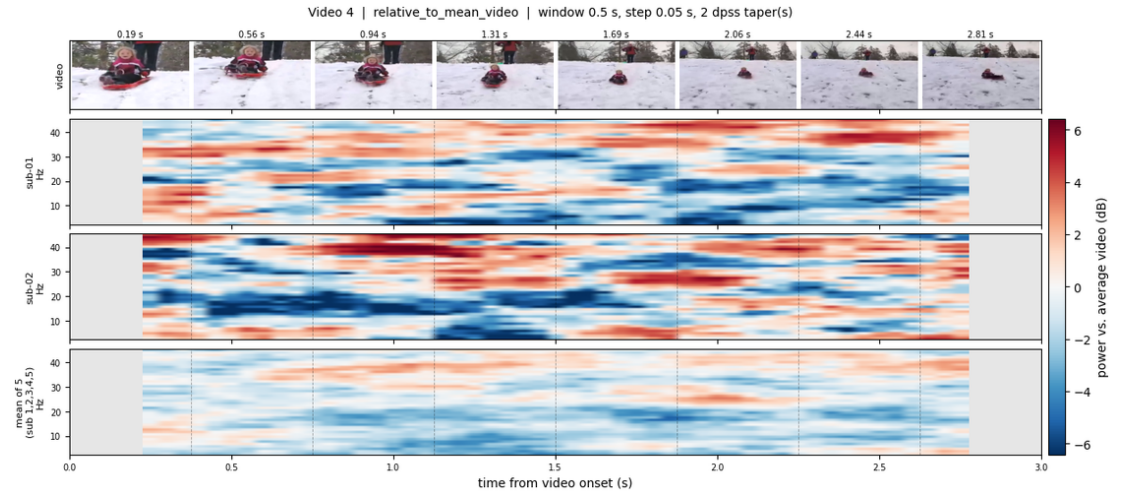
The exploratory mean spectrum also suggested attenuation above approximately 45 Hz. This observation alone does not establish the acquisition or preprocessing filter specification.

2.1.2 Frequency-Band Analysis

We assessed whether recordings of the same video were more similar across subjects than recordings of different videos within selected frequency bands. This measures cross-subject stimulus-specific similarity rather than the total information available in each band.



(a) Power normalised by each subject's mean training spectrum.



(b) Power normalised by each subject's time-frequency reference.

Figure 1: Time-frequency representations for training video 4. Each panel shows subjects 01 and 02 alongside a cross-subject summary over the five subjects with available recordings. The two panels use different subject-specific normalisation schemes.

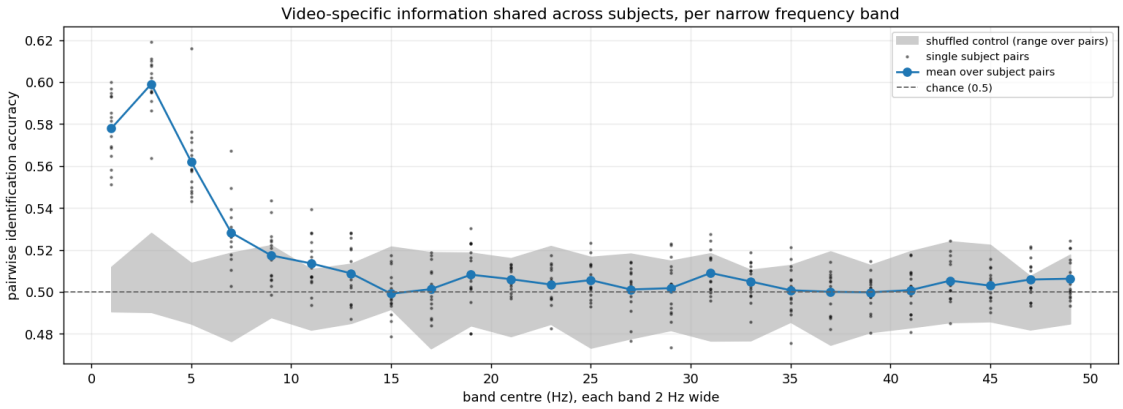


Figure 2: Cross-subject pairwise discrimination by frequency band. For each 2 Hz band, the metric is the proportion of comparisons in which recordings of the same video are more similar than recordings of different videos. The nominal chance level is 0.5. This metric is not candidate-set retrieval accuracy.

Method. We applied band-pass filtering and subtracted each subject's mean response across recordings. Mean-response subtraction reduces activity shared across stimuli; its effect is not limited to the onset response. For each pair of subjects, we selected the training videos with recordings available for both subjects.

Using correlation-based similarity, we compared a recording from one subject with the other subject's recording of the same video and with recordings of different videos. Pairwise discrimination accuracy was defined as the proportion of comparisons in which the same-video similarity exceeded the different-video similarity. This is distinct from Top-1 accuracy over a full candidate set. We used a nominal chance level of 0.5 and a video-label-shuffling control as an empirical reference. The analysis was repeated for 2 Hz-wide frequency bands spanning 0–50 Hz.

Results and Interpretation. The exploratory comparison suggested that same-video similarity was most evident in the lower-frequency bands and approached the nominal chance level from approximately 14 Hz (Figure 2). This result concerns cross-subject structure detected by the chosen similarity measure; it does not establish the absence of useful subject-specific or nonlinear information at higher frequencies.

These observations motivated an emphasis on low-frequency EEG in subsequent models. The final internal ensemble uses the configured band parameters 0.2–12 Hz, selected during later model development (Section 2.8). Its upper raised-cosine transition extends from 10 to 14 Hz. The exploratory analysis does not establish this configuration as an optimal frequency range for EEG retrieval.

2.2 Improved Baseline: Video-to-EEG Encoding with R(2+1)D-18

We started with a simple improvement of the baseline approach given with the task itself. It predicts the EEG response to each candidate video and ranks the 102 candidates by the Pearson correlation between measured and predicted EEG.

Instead of the frame-wise AlexNet features we use the frozen spatio-temporal video network R(2+1)D-18 (6), pretrained on Kinetics-400 (7) (torchvision, `R2Plus1D_18_Weights.KINETICS400_V1`, weights `r2plus1d_18-91a641e6.pth`). Activations of layers 1–4 are pooled to at most four time steps and a 2×2 spatial grid, standardised and reduced by PCA. Ridge regression then maps them to the EEG, with the penalty chosen by leave-one-out cross-validation.

From the data analysis, we saw that the videos created different frequencies in the EEG data. Hence, we wanted to try using the time-frequency information instead of the raw EEG or use both. We further saw, that most activity is at the beginning of the video. Hence, we evaluated the results when limiting the EEG data to different time windows.

We used a grid search to try different configuration including the target (raw EEG, time–frequency power, or both) and different time windows.

Results. Each subject gets the configuration with the best top-5 accuracy on 20 % held-out training videos. Test data are used only once, for the final evaluation.

On the test set of subject 1, the configuration chosen on validation reaches 7.8 % top-1 and 18.6 % top-5 accuracy (median rank 17.5). The AlexNet baseline reached 3.9 %, 11.8 % and 25.5 (chance: 1.0 %, 4.9 %, 51.5).

The validation runs show that the video information lies in the time-locked evoked response: spectral power adds nothing over the raw EEG. Further, the best configuration used a window of $(0, 0.7s)$. This means, that the beginning of the video actually contains more information and it was wrong to assume that the high activity at the start is only non video specific noise. In our further trainings and tests, we could confirm this.

2.3 Video Feature Extraction

In an early video-to-EEG encoding approach, we used timestamped stimulus features to predict each subject's EEG response. Separate time-lagged ridge encoders were fitted for each subject, and candidates were ranked using Pearson-correlation-based similarity between predicted and recorded EEG. We evaluated simple image and motion statistics alongside pretrained visual representations. This subsection describes the early ridge approach, whose temporal grid and feature processing differ from the final dual-encoder system.

Temporal Alignment. Features were aligned using decoded presentation timestamps. EEG was downsampled from 500 to 100 Hz, and the default encoder used stimulus features at lags from 0 to 300 ms relative to each EEG sample. Thus, the lag range included both the current and preceding stimulus features. The initial period requiring unavailable pre-video history was excluded.

Frame Difference. Frame difference was defined as the mean absolute difference between consecutive grayscale frames. It measures image change without distinguishing object motion, camera motion, or changes in illumination. The first frame was marked as unavailable because no preceding frame existed. The exploratory result did not establish a clear retrieval advantage for this feature alone.

Optical Flow. Dense optical flow was estimated using the Farnebäck algorithm. Frames were resized to a maximum width of 640 pixels before flow estimation. Displacements were divided by the elapsed time between the corresponding frames, yielding image-plane velocities in processed-

image pixels per second. These velocities depend on image scaling and are not physical object velocities.

The features comprised mean and 90th-percentile speed, mean horizontal and vertical velocity, and mean speed within nine image regions. Camera motion was not compensated. The motion-only encoder achieved 1.72% Top-1 accuracy and 7.60% Top-5 accuracy across the four DEV subjects, compared with uniform-random expectations of approximately 0.98% and 4.90%, respectively. These exploratory scores do not establish a statistically reliable improvement.

Brightness. Brightness was represented by the mean grayscale intensity of each frame, scaled to $[0, 1]$. Across the four DEV subjects, the brightness-only encoder yielded 5/408 correct Top-1 predictions (1.23%). One-to-one assignment increased the reported total to 7/408 (1.72%). Uniform-random Top-1 rankings have an expected 4/408 correct predictions (0.98%). The small differences in observed counts do not by themselves establish reliable retrieval from global brightness.

Contrast. Contrast was represented by the within-frame standard deviation of grayscale intensities. The contrast-only encoder yielded 2/408 correct Top-1 predictions (0.49%), increasing to 3/408 (0.74%) after one-to-one assignment. Both observed totals were below the uniform-random expectation of four correct predictions. These results provide no clear evidence of useful retrieval from this global contrast descriptor in the tested setup.

Learned Visual Features. We also extracted frozen pretrained AlexNet activations from each decoded frame, followed by spatial pooling and dimensionality reduction fitted on training data. The available AlexNet-only results covered three subjects and yielded 4/306 correct Top-1 predictions (1.31%). This smaller evaluation set should be distinguished from the four-subject results above.

The reported combination of scalar audiovisual features and AlexNet achieved 1.72% Top-1 accuracy and 6.86% Top-5 accuracy. Because this representation also included audio, it is not a video-only ablation. Assessing the contribution of feature combination requires evaluation on the same subjects and recordings; the reported aggregates alone do not establish an improvement.

2.4 Audio Feature Extraction

This subsection describes early audio-to-EEG ridge experiments. Their window lengths, spectral features, and decoding procedures should be distinguished from the final audio feature pipelines.

RMS Features and Audio Timing. Audio amplitude was represented by root-mean-square (RMS) values computed over 25 ms windows with a 10 ms step. For a window of L samples and C audio channels, the feature was

$$a_m = \left[\frac{1}{CL} \sum_{c=1}^{C} \sum_{\ell=0}^{L-1} x_c[n_m + \ell]^2 \right]^{1/2}, \tag{1}$$

where $x_c[n]$ denotes the decoded audio samples in channel c and n_m is the start of window m. Squared samples were thus averaged over channels and time before taking the square root.+Features were timestamped at window centres on the decoded audio timeline. Preserving audiovisual offsets additionally requires a common time reference for the audio and video streams.

During development, frame-duration handling was corrected after observing discrepancies between decoded sample counts and reported presentation intervals in some AAC audio. Genuine

gaps were retained, and interpolation was restricted to contiguous segments. Videos without an audio stream were represented as silence in this early feature pipeline.

The RMS-only encoder achieved 19/408 correct Top-1 predictions (4.66%) and 59/408 correct Top-5 predictions (14.46%). These are exploratory DEV results from subjects 01–04. They motivated further investigation of audio features; they do not by themselves isolate a modality effect relative to the earlier visual models.

Richer Audio Features and Model Selection. A subsequent experiment added eight timestamped log-mel energy bands and used nested cross-validation for subject-specific configuration selection. The common frequency range was chosen with reference to the lowest available audio sample rate.

Benefits varied across subjects. In an early experiment using a 300 ms delay setting, the training-selected models achieved 22/408 correct Top-1 predictions after one-to-one assignment (5.39%). Independent rankings achieved 15.93% Top-5 accuracy. These figures refer to different ranking procedures and do not describe a single output ranking.

The reported assigned Top-1 gain of 0.74 percentage points was relative to the RMS reference evaluated within that experiment, not the earlier result of 25/408. The reported approximate conditional 95% interval was $[-0.49, 1.96]$ percentage points. It included zero, so the reported comparison did not establish a reliable gain.

Later delay-range and channel-weighting experiments also produced mixed results. These findings support using RMS as a reference when assessing extensions, rather than a general conclusion that richer audio representations are unhelpful. All reported 102-candidate results concern DEV subjects 01–04. Repeated inspection and tuning prevent treating them as an untouched evaluation benchmark.

2.5 Clustering and Early Embedding Retrieval

In an exploratory clustering approach, visual representations were reduced using principal component analysis (PCA) and grouped using k-means. A separate model for each subject predicted these cluster assignments from EEG features. The search varied the video and EEG PCA dimensions and the number of clusters.

The reported selected configuration used 192 video PCA dimensions, 96 EEG PCA dimensions, and seven clusters. Its selection score was 0.051 and its mean normalised cluster lift was 0.0678. The lift expressed improvement over a uniform-random reference relative to the maximum possible improvement; the selection score additionally ccounted for variation across subjects. These scores are not candidate-level retrieval accuracies.

Clustering was not retained in the final retrieval pipeline. A cluster prediction does not by itself distinguish videos assigned to the same cluster, whereas the task requires a complete ranking of individual candidates. We therefore also investigated a direct mapping from EEG to visual embeddings. Cluster lift and retrieval metrics evaluate different outcomes and cannot establish comparative performance without a common candidate-ranking evaluation.

For early embedding retrieval, configuration selection rotated over DEV subjects 01–04: configurations were selected using three subjects and assessed on the remaining subject. The same configuration was selected in all four rounds: full-video CLIP features, 144 video PCA dimensions, 192 EEG PCA dimensions, and 384 hidden units. The EEG mapping for each assessed subject was still fitted using that subject's own training recordings. This procedure evaluates configuration transfer across subjects, not generalisation to a subject with no training EEG.

Mean Top-1 and Top-5 accuracies were reported as 2.97% and 10.32%, respectively, with averaging over subjects and individual training seeds. For 102 candidates, the uniform-random expectations are approximately 0.98% and 4.90%. Reuse of the same DEV subjects in earlier experiments limits the independence of these results.

For subjects 05–06, the selected configuration was used to train eight seeds per subject, whose predictions were ensembled. A one-to-one assignment enforced unique first-ranked candidates within each subject's 102 recordings. The resulting historical files contained complete 102-candidate rankings. Their construction is an intermediate prediction route, not the final v7 submission. Blind-subject accuracy is unknown, and the single-seed validation results do not directly measure the performance of this ensemble.

2.6 Predicting Video Features from the EEG Data

Instead of first predicting EEG responses from candidate videos, an EEG decoder estimated features extracted by a frozen visual encoder. Retrieval then compared this prediction with the representations of the 102 candidate videos.

Method and Experimental Setup. Each recording contained 17 EEG channels sampled at 500 Hz over three seconds. The signals were downsampled to 100 Hz using anti-aliased polyphase resampling. We compared the complete recording with an early window covering the first 600 ms. Visual targets were extracted using frozen DINOv2 ViT-L/14 (3) and V-JEPA 2 ViT-L/16 encoders. Two representations were evaluated: a global average over the clip and a temporal representation formed by concatenating averages from three consecutive temporal bins. Feature standardization and principal component analysis (PCA) reduced the target dimensionality; the extended search considered 16, 32 and 64 dimensions. The resulting target vectors were normalized.

Two decoder families were compared. Subject-specific ridge regression provided a regularized linear baseline. Shared neural decoders combined EEG processing with a learned subject embedding. These included a generic temporal convolutional network and a compact EEGNet-inspired architecture using temporal, depthwise spatial and separable temporal convolutions. The latter was a custom adaptation rather than a reproduction of the published EEGNet architecture. Neural training minimized the squared distance between normalized predictions and targets. An additional contrastive objective encouraged the correct video to receive a higher similarity than competing videos:

$$\mathcal{L} = \mathcal{L}_{\text{regression}} + \lambda \mathcal{L}_{\text{contrastive}}. \tag{2}$$

Recordings of the same video across subjects were treated as positive pairs rather than false negatives. Candidate ranking used cosine similarity after subtracting the training-derived target mean.

Evaluation used five fixed splits, each containing 102 held-out training video identities. Held-out identities were excluded across all subjects. Adaptive EEG preprocessing, feature scaling and PCA were fitted only on the training portion of each split. These folds also selected hyperparameters and neural stopping epochs; consequently, the reported scores describe model-selection performance rather than an independent final test.

Retrieval Performance. The highest reported Top-1 accuracy was 7.29%. This configuration used the compact EEGNet-inspired decoder, a 64-dimensional temporal V-JEPA 2 target, the

Configuration	Top-1 (%)	Top-5 (%)	MRR
Random ranking, expected	0.98	4.90	0.0510
V-JEPA 2: ridge, temporal	6.75	23.69	0.1670
V-JEPA 2: ridge, global	6.72	23.83	0.1680
V-JEPA 2: neural, temporal	7.29	22.03	0.1632
V-JEPA 2: neural, global	7.18	21.35	0.1610
DINOv2: neural, best reported	5.17	18.26	0.1361

Table 2: Selected results of the early standalone visual-feature approach on held-out training videos. Neural metrics are averages over training seeds, not measurements of an ensemble. The same validation splits informed hyperparameters and stopping epochs.

early EEG window, band-pass filtering with OAS spatial whitening, and the combined regression and contrastive objective. Although approximately 7.4 times the random-ranking expectation, its absolute accuracy remained low.

The strongest ridge configuration reached 6.75% Top-1 accuracy using the same target family, dimensionality, pooling and early window. The neural model therefore improved Top-1 by approximately 0.54 percentage points. However, ridge achieved higher Top-5 accuracy and MRR. The neural model's advantage was thus specific to the primary selection metric, rather than a consistent improvement throughout the ranking.

Interpretation of the Retrieval Results. The reported point estimates were above the random-ranking expectation for both linear and neural decoders. They provide exploratory evidence of correspondence between EEG and the selected visual targets under the tested protocol, rather than an independent estimate of blind generalisation performance.

V-JEPA 2 targets produced stronger reported configurations than DINOv2 targets. This was consistent with the motivation for using a video representation rather than relying only on frame features. Nevertheless, the best configurations differed in other settings, so this comparison does not isolate encoder choice as the sole cause.

Preprocessing also mattered. For the selected temporal ridge setup, oracle approximating shrinkage (OAS) whitening with band-pass filtering increased Top-1 from a reported matched untreated baseline of 3.23% to 6.75%, a gain of approximately 3.5 percentage points. The corresponding neural temporal configuration improved from 6.65% without cleaning to 7.29% with whitening. These comparisons concern preprocessing recipes. The ridge comparison changes filtering and whitening together and does not isolate either operation as the sole cause of the difference.

The combined objective had the higher point estimate in the reported neural comparison. With the same listed temporal target, whitening, window and optimiser settings, regression alone achieved 6.37% Top-1, compared with 7.29% for the combined objective. This is consistent with a benefit from the contrastive term in that comparison, but does not by itself quantify uncertainty or establish a general advantage across configurations.

Limitations and Potential Bottlenecks. The principal limitation was low absolute retrieval accuracy. A trainable nonlinear decoder only modestly exceeded ridge at rank one, while ridge retained better Top-5 and MRR. The tested nonlinear architecture therefore did not yield a consistent gain across ranking metrics.

The target representation was another plausible limitation. Both pooling strategies compressed a three-second video into a single target vector, with the temporal variant preserving only three coarse bins. They did not explicitly model fine-grained correspondence between EEG responses and stimulus changes. The reported temporal and global neural configurations differed by approximately 0.11 percentage points in Top-1 accuracy. This small difference between selected configurations does not establish a causal benefit from three-bin temporal pooling.

PCA also retained directions of visual feature variance, rather than directions selected for predictability from EEG. Some retained visual information could therefore have been difficult to decode, while weaker but EEG-relevant directions could have been discarded. This remains a methodological explanation, not a mechanism established by the reported experiments.

Cleaning was not uniformly beneficial. For example, the selected ICA configuration underperformed its matched untreated ridge baseline by approximately 0.38 percentage points. Without dedicated EOG channels or documented electrode geometry, artifact identification was limited. Whitening should consequently be interpreted as a useful spatial transformation, not evidence that artifacts were successfully removed.

Finally, the best configurations used the first 600 ms, but this does not establish that later EEG contained no useful information. Filtering and resampling operated on complete trials before cropping, making this an offline window comparison rather than a causal latency analysis. Likewise, small variation across training seeds does not quantify uncertainty across unseen videos or correct the optimism introduced by searching many configurations.

The original concept therefore provided an exploratory visual-decoding baseline with point estimates above chance and identified candidate choices for preprocessing and training. Its performance nevertheless remained insufficient for reliable video identification, and the available pre-integration results did not establish blind-test generalization.

2.7 Data Augmentation in Early Encoding and Decoding Models

The reported dataset inventory comprised 1,000 training videos and 5,650 EEG–video pairs from six subjects. We evaluated noisy copies of EEG targets, additional video views, and online EEG-input perturbations. These operations increased the number or variation of training examples without adding independent observed stimuli or EEG recordings. Video identities were split globally across subjects; only training examples were augmented, and validation used original recordings. Comparisons used controls from the same experiment.

Noisy Targets in the Linear Encoding Model. Ridge regression mapped visual features to EEG targets. Each training recording was duplicated with zero-mean Gaussian noise whose standard deviation was 2% of the channel standard deviation. The reported training count increased from 4,516 to 9,032 rows. With EVA02-CLIP features, Top-1 and Top-5 totals remained 19 and 85 out of 1,134 validation pairs; subject-macro MRR changed from 0.064102 to 0.064095. For AlexNet, the reported Top-1, Top-5 and median rank were unchanged for every subject.

For fixed inputs x_i, targets y_i, duplicated targets $\tilde{y}_i = y_i + \varepsilon_i$, and fixed α under the stated sample-count-scaled penalty, the objective satisfies $\sum_{i=1}^{n} \left(\|y_i - Wx_i\|^2 + \|\tilde{y}_i - Wx_i\|^2\right) + 2n\alpha\|W\|_F^2$
$= 2\left[\sum_{i=1}^{n} \|\bar{y}_i - Wx_i\|^2 + n\alpha\|W\|_F^2\right] + \frac{1}{2}\sum_{i=1}^{n} \|\varepsilon_i\|^2, \qquad \bar{y}_i = y_i + \frac{\varepsilon_i}{2}$. The additional term is independent of W. Thus, this duplication scheme is equivalent to ridge fitting on averaged targets, not to obtaining independent new observations. For zero-mean noise conditional on fixed inputs and

Variant	Macro MRR	Δ	Folds > 0
Control	0.1155	—	—
Control replicate, different seeds	0.1145	-0.0010	2/3
Gaussian noise, $\sigma = 0.1$	0.1144	-0.0011	1/3
Gaussian noise, $\sigma = 0.25$	0.1124	-0.0031	0/3
Gaussian noise, $\sigma = 0.5$	0.1063	-0.0092	0/3
Channel dropout, $p = 0.1$	0.1031	-0.0124	0/3
Amplitude scaling, $\sigma = 0.1$	0.1144	-0.0011	1/3
Temporal shift, 0–40 ms	0.1118	-0.0037	1/3
Time mask, ≤ 100 ms	0.1175	$+0.0020$	3/3

Table 3: Reported online-augmentation results for five-seed ensembles, averaged over three folds. Differences are calculated from the displayed macro-MRR values relative to 0.1155. The fold counts are retained from the original report.

fixed regularisation, the ridge coefficients are unchanged in expectation; a finite noise draw can still change predictions and rankings. This argument does not prove that every augmentation of a linear model is ineffective.

Video-View Augmentation. We also retained the EEG targets and added three video views: two crops retaining 95% of image width and height and one mild brightness/contrast modification. Training rows were quadrupled. The reported comparison did not show a convincing gain, and the re-tuned configuration performed worse.

Small additive input noise can act as a regulariser under suitable assumptions (**?**). This does not establish an exact equivalence for the finite image transformations used here. Preserving the original EEG target for an edited video is itself an augmentation assumption: those edited stimuli were not presented to the subjects.

Online EEG Augmentation in the Contrastive Decoder. For an early contrastive decoder, normalised training EEG was perturbed online with Gaussian noise, channel dropout, per-channel amplitude scaling, temporal shifts up to 40 ms, or masking of a time segment up to 100 ms. Selection used three disjoint video validation folds and five-seed ensembles at selected fixed epochs. A control replicate used five different seeds. The stated adoption rule required a mean macro-MRR gain of at least 0.003 and a positive gain in every fold; this is an empirical selection rule, not a statistical significance threshold.

No variant met the stated adoption rule. Time masking improved macro MRR by 0.0020 and was positive in all three folds, but its mean gain was below 0.003. The single control replicate differed by -0.0010; one replicate does not estimate the full distribution of seed effects and does not establish whether the masking gain lies within that distribution. Increasing the tested Gaussian noise strength lowered the displayed mean MRR. Channel dropout and temporal shifts also had lower means in this comparison.

These findings do not exclude overfitting as a limitation. Perturbations may change useful temporal or spatial structure, and their effects can depend on strength, optimisation and model family. A fixed electrode montage does not by itself make channel dropout universally inappropriate. Physiological explanations for the observed changes were not isolated by these comparisons.

Interpretation. A separate reported comparison used a control macro MRR of 0.117. Adding an audio branch, changing the EEG encoder to a spatiotemporal convolutional architecture, and combining both changes gave reported gains of 0.020, 0.015, and 0.036, respectively, with positive gains in every validation fold. The control value differs from the 0.1155 in Table 3; these comparisons should not be treated as one identical run without the original records.

The tested early augmentations did not meet the chosen adoption criterion. This limits conclusions to those models and settings, not augmentation in general. In particular, these experiments do not evaluate the later matched-splice procedure (Section 3.4). Training-row counts alone do not measure the benefit of augmentation; comparisons should evaluate the deployed ensemble and report uncertainty across relevant sources of variation.

2.8 EEG Preprocessing

Preprocessing differed across model families. For the final internal ensemble, EEG sampled at 500 Hz is filtered using a zero-phase FFT filter with configured band parameters of 0.2 and 12 Hz and then subsampled by a factor of ten. The filter uses raised-cosine transitions and 0.5 s of reflection padding on each side. Its upper transition extends from 10 to 14 Hz; the configured band edges are not ideal rectangular cutoffs. Each three-second recording becomes a 150×17 time-by-channel array.

For each subject and each set separately, let u_{ict} denote the filtered, subsampled EEG for trial i, channel c, and time t. The internal transformation is $\mu_{ct} = \frac{1}{N}\sum_{i=1}^{N} u_{ict}$, $\sigma_c = \left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}(u_{ict} - \mu_{ct})^2\right]^{1/2}$, $z_{ict} = \frac{u_{ict}-\mu_{ct}}{\sigma_c+10^{-8}}$, where $T = 150$ and N is the number of recordings in that set. Mean-response subtraction acts at every channel and time point, not only at stimulus onset. Channel scaling uses the centred recordings and a population standard deviation over trials and time.

Training and test statistics are computed separately. Test-set normalisation uses all unlabelled recordings of that subject, so the transformed value of one query depends on the other queries. This is transductive preprocessing; it uses no EEG–video correspondence and does not fit encoder weights on test recordings. The zero-phase filtering and full-recording transformation are offline operations, not an end-to-end causal preprocessing pipeline.

The supplied audio and visual families instead apply stored training-fitted normalisation parameters. The audio family uses its own 50 Hz processing, training mean response and channel scales. The visual transformer uses FIR decimation to 100 Hz, a short EEG window, training-fitted Ledoit–Wolf whitening, and channel scaling. These paths are not interchangeable with the internal transformation.

Spatial covariance whitening was also explored in intermediate linear models. Those experiments do not make whitening a common step of all final internal models. Preprocessing descriptions and reported comparisons must refer to the model family actually used.

Preprocessing is part of the model specification. Its fitted statistics, dependence on other test queries, and offline temporal operations must be stated alongside the supervised training boundary.

2.9 Label-Dependent Test Templates and the Training-Data Reset

An exploratory route used labelled test EEG responses, or representations derived from them, as candidate-specific templates or supervised targets. A linear EEG encoder in that route achieved

a reported 62.25% DEV Top-1 accuracy. Because the information path included the candidate correspondence of test responses, this score does not measure retrieval under the final training-data protocol. We excluded the route from the submission and restricted subsequent supervised fitting to official training recordings. This boundary is distinct from the label-free transductive preprocessing described in Section 2.8.

The first replacement baseline achieved 1.96% Top-1 accuracy, 8.82% Top-5 accuracy, and MRR 0.0738. These figures and the template-based result use different information protocols; their difference should not be interpreted as an isolated architectural effect.

A subsequent spectrotemporal reconstruction combined spatial filtering, temporal convolution, event-related response windows, bandpower branches, and visual or motion targets. A nested subject-wise fusion-selection procedure reported 5.64% Top-1 accuracy. After refitting on all training recordings, DEV results were 6.37% Top-1, 21.81% Top-5, and MRR 0.1656. These are development measurements, not blind-subject performance.

The label-dependent route, the early replacement baselines, and the final v7 model are separate stages. Their historical implementations and prediction files are not included in the focused final release. The supplied final code implements the later training-recording workflow and does not provide a reproduction of these historical scores.

Development scores must be interpreted together with their information paths. Candidate-specific test-response templates cannot serve as evidence for a system whose inference must not depend on those correspondences. Excluding that route restored the intended boundary for subsequent supervised fitting; independent evaluation remained necessary to assess generalisation.

3. Multimodal temporal encoding

The temporal approach replaces static stimulus targets with aligned audiovisual feature sequences. The final internal model uses the full three-second recording and a nominal EEG passband of 0.2–12 Hz. These choices were selected within the tested configurations and do not establish that informative activity is confined to this frequency range or requires the complete recording in every model.

3.1 Linear temporal-response models

A temporal response function (TRF) maps time-lagged stimulus features to EEG, representing delayed responses to stimulus events. The historical development record reports DEV Top-1 accuracies of 10.05% for an initial low-level audiovisual TRF, 9.31% for its audio-only counterpart and 2.70% for its visual-only counterpart. Subsequent configurations incorporating spatial whitening and Whisper features reportedly exceeded 20%, including a 21.32% Top-1 result.

These preliminary results motivated the nonlinear temporal model. A TRF was also evaluated as an additional score family; the reported gains in Top-5 accuracy and MRR coincided with lower Top-1 accuracy, and this family was excluded from the final ensemble. The corresponding historical code, lag specifications and score matrices are absent from the supplied release; these figures therefore remain development milestones rather than independently reproduced comparisons.

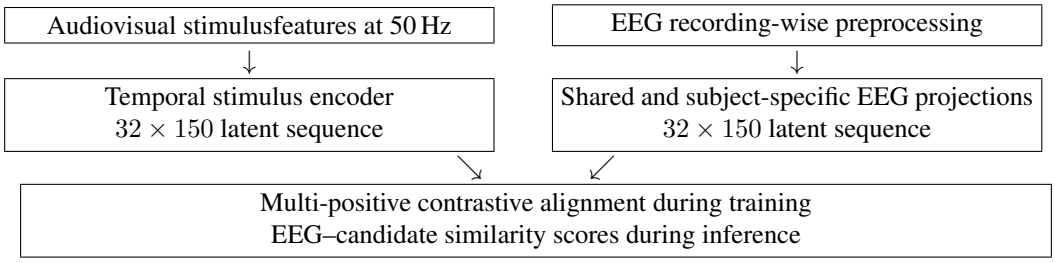


Figure 3: The internal temporal dual encoder. Available training responses to the same video share a positive stimulus identity. Both branches operate offline. Seed ensembles, feature families and the supplied audio and visual models are subsequently combined at score level.

3.2 Dual-encoder architecture

The internal model jointly learns a stimulus encoder and an EEG encoder (Figure 3). Stimulus features are represented on a 50 Hz grid with 150 time steps. An input projection maps the feature vectors to 128 channels. Four residual temporal convolution blocks use kernel size five, dilations 1, 2, 4 and 8, gated linear units, group normalisation and dropout. A pointwise output projection produces 32 latent channels.

The convolutions use left padding, but group normalisation aggregates statistics over channels and the complete temporal axis. Consequently, the stimulus encoder itself is not temporally causal. The EEG encoder adds shared and subject-specific $17 \rightarrow 32$ spatial projections and applies a depthwise temporal convolution with kernel size nine and symmetric padding. Both branches return a 32×150 latent sequence. Subject-specific parameters are learned from the training recordings of all six subjects; this is a known-subject retrieval setting.

Each latent channel is centred and divided by its temporal standard deviation plus 10^{-6}. The flattened representation is divided by $\sqrt{32 \cdot 150}$. Dot products between the resulting vectors define correlation-like similarities. At inference, the mean stimulus vector over the candidate bank is subtracted after this transformation, without subsequent renormalisation. This operation uses unlabelled stimulus features. The learned temperature scales training logits and is omitted from inference scores. The v1 ensemble, comprising 16 seeds, was reported to achieve 34.56% DEV Top-1 and 66.42% Top-5 accuracy; its historical checkpoints and score matrices are not included in the final release.

3.3 Multi-positive training

Each unaugmented batch contains 16 distinct training videos and all available training trials for those videos across subjects. Thus, each EEG query has one target video, whereas a video can have several positive EEG observations. The number of EEG trials in a batch depends on the available recordings.

Let a_{iv} denote the temperature-scaled similarity between EEG trial i and video v, let $v(i)$ be the target video of trial i, and let $P(v)$ contain all matching trials in the batch. For the unaugmented batch, with N_E trials and N_V videos, the directional losses are $\mathrm{L}_{E \to V} = -\frac{1}{N_E} \sum_{i=1}^{N_E} \log \frac{\exp(a_{i,v(i)})}{\sum_{u=1}^{N_V} \exp(a_{iu})}$,
$\mathcal{L}_{V \to E} = -\frac{1}{N_V} \sum_{v=1}^{N_V} \frac{1}{|P(v)|} \sum_{i \in P(v)} \log \frac{\exp(a_{iv})}{\sum_{j=1}^{N_E} \exp(a_{jv})}$. The objective is their arithmetic mean. The

reverse direction averages the negative log-probabilities of individual positive trials; it does not take the negative logarithm of their summed probability. The inverse temperature is learned, initialised to $1/0.07$ and capped at 100.

Recorded training-video holdout sweeps contain ensemble MRR values of 0.257391 for an earlier recipe and 0.345861 for a subsequent multi-positive recipe. These are sequential development configurations: batch construction, noise, dropout and training schedules changed during this phase. The difference cannot be attributed exclusively to the loss or batch construction. The historical 24-seed v2 refit was reported to reach 54.66% DEV Top-1, 82.60% Top-5 and MRR 0.6687; its DEV score matrices are absent from the supplied release.

The frozen internal recipe uses AdamW with maximum learning rate 0.002, weight decay 0.05, input and hidden dropout of 0.5, no additive EEG noise, and gradient clipping with maximum norm five. Training stops after epoch 22 of a OneCycle schedule configured for 30 epochs. This fixed stopping point is distinct from validation-based early stopping. These settings describe the released configuration and are not individually established optima.

Interpretation. Grouping responses by stimulus makes their shared identity explicit in the objective. The development results support the resulting training recipe, while leaving the contribution of individual changes unresolved.

3.4 Matched temporal splice augmentation

The augmentation joins the beginning of one stimulus feature sequence to the end of another. Corresponding EEG segments are taken from the same subject and joined at the same cut. In each batch, a permutation of the 16 video identities defines the pairs. A single cut index is sampled uniformly from $\{30, \ldots, 120\}$ and shared by that batch's synthetic pairs, corresponding to 0.6–2.4 s in 20 ms steps. The final recipe attempts one splice set per batch. Pairs with identical source videos are excluded from synthetic EEG construction; other pairs require both recordings from the same subject.

Splicing is performed on precomputed features and preprocessed EEG. It produces matched synthetic training pairs, but does not simulate the physiological response to an edited audiovisual stimulus: feature context, neural response delays and preprocessing can cross the cut. In six-seed evaluations, the stored non-spliced configuration stopped at epoch 15 of a 20-epoch schedule yields holdout MRR 0.349203. A recorded spliced configuration with the same schedule and stopping point reaches 0.357746. A subsequent 30-epoch schedule stopped at epoch 22 reaches 0.364315 in the stored base-family scores. These sequential results support the tested recipe without isolating the contribution of augmentation from changes in schedule and experimental run. The recorded two-cut variant reached 0.364111, while two splice sets reached 0.354865 under the 30-epoch schedule stopped at epoch 22; neither exceeded the corresponding single-cut, single-set result.

The frozen implementation retains all pseudo-stimulus columns whenever at least one synthetic EEG pair is available. Fixed points in the permutation or unavailable same-subject recordings can therefore leave columns without positive EEG observations. These columns remain in the EEG-to-video softmax denominator. A fixed point can duplicate an original stimulus and introduce a false negative. In the reverse loss, an empty column contributes zero but remains in the mean over stimulus columns. These behaviours are retained in the released result; the effect of filtering such columns has not been measured.

Family	Feature composition
G	Low-level audiovisual descriptors; Whisper PCA-32 per hidden-state index; DINOv2 PCA-16.
O	G with an additional 64-band log-mel representation and 64 spectral flux features.
D	Low-level audiovisual descriptors; Whisper PCA-64 per hidden-state index; DINOv2 PCA-16; CLIP PCA-16.
H	Low-level visual descriptors; Whisper PCA-64 per hidden-state index; DINOv2 PCA-16. Only the low-level audio columns are omitted.
I	Low-level audiovisual descriptors; Whisper PCA-64 per hidden-state index; 64-band log-mel and 64 spectral flux features; DINOv2 PCA-16.

Table 4: Feature composition of the five internal families. PCA dimensions refer to retained components. All families use 150 time steps and include Whisper audio features; H is therefore also multimodal.

Interpretation. Matched splicing increases training variation without additional recordings. Its empirical contribution is specific to the tested recipe, and both the synthetic-pair assumption and positive-label construction limit its interpretation.

3.5 Feature families and seed ensembles

The internal families combine low-level descriptors with Whisper-base hidden states at indices 2, 3 and 4 (2) and DINOv2-base image features (3). One family additionally uses CLIP ViT-B/32 features (1). PCA projections and feature standardisation are fitted on the training videos and applied unchanged to candidate videos. An exception precedes these transformations: in the low-level audio branch, clips without detected audio receive mel and envelope values averaged over audible clips in the complete loaded stimulus bank, including the 102 unlabelled candidate videos. This is a transductive feature-construction step. The separate audio-rich branch uses audible training clips for its replacement values.

Two greedy forward searches on training-video holdout scores selected G/H/I and G/O/D/H/HuBERT, respectively. The final G/O/D/H/I ensemble was curated from these searches. HuBERT was omitted because its addition increased validation MRR by approximately 0.0008 while requiring another model dependency. Removing I from the locked five-family combination increased validation MRR from 0.399629 to 0.400127. Accordingly, neither the curated family set nor equal family weights are established as optimal. These comparisons use six seeds per internal family and a repeatedly reused validation split.

The final internal ensemble contains twelve seeds per family, giving 60 models. Each seed's score matrix is standardised within query rows before averaging within its family. The resulting family matrices are again standardised within rows and averaged with equal weights. The historical v4 and v5 DEV Top-1 accuracies were reported as 58.58% and 58.82%, respectively, a difference corresponding to one net additional correct query among 408. The supplied release does not contain the historical v4 DEV scores needed to reproduce that comparison.

4. Video and Audio Transformers

The final system adds two independently, self-trained transformers as separate score families. Both are small transformers trained with the help of large public EEG datasets, because transformers trained only on the challenge recordings had remained at chance level. Standalone, they were selected on five folds of 102 held-out training videos (four seeds, all six subjects, training-only normalisation). We assessed their contribution to the final system on held-out training-video scores rather than assuming that a more elaborate encoder would help.

4.1 Audio dual encoder

The audio model is the temporal dual encoder of Section 3, which follows CORTIVA (8), restricted to sound. Its input per 20-ms step is the loudness envelope, Whisper's 80-band log-mel spectrum and Whisper-small layer-12 features reduced by PCA to 64 dimensions. The EEG side adds a two-layer transformer over time (model dimension 64, four heads). Its output projection starts at zero, so training begins from the convolutional encoder. We pretrained it on the ICASSP 2023 Auditory EEG Challenge data (9; 4) (71 listeners, approximately 120 h of speech EEG, restricted to our 17 channels and cut into 3-s windows). On held-out speech it reached 64.4% Top-5 among 102 windows. This pretraining did not transfer to our videos: fine-tuned and from-scratch training scored equally (32.7% vs. 33.0% validation Top-5), and the transformer layers did not beat the purely convolutional encoder (33.1%). The supplied model is therefore trained from scratch on the challenge recordings. Its only dependency on SparrKULee is the Whisper PCA basis, which was fitted on the speech audio. Standalone, it reached 35.05% Top-1 and 60.54% Top-5 on the test set.

We tested feature concatenation, reuse of its EEG encoder and independent score fusion. The first two did not provide a stable holdout gain. Independent fusion did: a 20% audio share increased training-video independent-ranking MRR from 0.399629 to 0.404841. The nearby tested weights were close, so 20% should not be interpreted as a precisely identified optimum. The corresponding v6 independent-ranking DEV result was 61.27% Top-1, 86.03% Top-5 and MRR 0.7267. Its 2.45-point Top-1 gain over v5 had a paired bootstrap interval touching zero.

4.2 Visual EEG transformer

The visual model maps 0–0.8 s of EEG at 100 Hz to the CLIP ViT-B/32 space (1). A per-subject 17×17 channel mixing is followed by 272 tokens (50-ms patches per channel, model dimension 128), four transformer layers with four heads and a linear head (0.87 M parameters). It was pretrained on 661,600 single trials of THINGS-EEG2 (5), which uses the same 17 channels in the same order and whose epochs end at 0.8 s. Pretraining used a symmetric InfoNCE loss against the CLIP embeddings of the shown images. The model was then fine-tuned on the challenge recordings against the mean CLIP embedding of the frames shown during 0–0.7 s, accounting for the approximately 0.1 s visual latency. EEG is whitened and scaled with training statistics. Here, pretraining was essential: fine-tuning reached 22.9% validation Top-5, against 7.8% for the same network trained from scratch. Without fine-tuning, the THINGS model stayed near chance. Standalone, the supplied model reached 10.29% Top-1 and 32.35% Top-5 on the test set. Applying it to later 0.8-s windows of the trial gave near-chance scores, so for this static target the decodable visual information lies in the onset response. This short-window family complements the internal full-duration models; it does not establish that an early window is universally preferable.

Keeping the audio share at 20%, the recorded weight grid selected a 10% visual share by independent-ranking holdout MRR, which rose to 0.417263. The final group shares are consequently 70% internal, 20% audio and 10% visual. The weight maximising assigned holdout Top-1 differed; we retained the declared MRR selection rule.

5. Global ranking and the final v7 system

5.1 Score fusion and one-to-one assignment

Let $Z(S)$ denote standardisation of each query row over the candidate scores, using the row mean and population standard deviation with 10^{-9} added to the denominator. The final score matrix for each subject is

$$S = 0.70Z(S_{\text{internal}}) + 0.20Z(S_{\text{audio}}) + 0.10Z(S_{\text{visual}}). \tag{3}$$

These are similarity scores, not calibrated probabilities. We apply a cross-domain similarity local scaling (CSLS) correction (**?**) to the fused matrix:

$$C_{iv} = 2S_{iv} - r_i^{(k)} - c_v^{(k)}, \tag{4}$$

where $r_i^{(k)}$ is the mean of the largest k entries in row i and $c_v^{(k)}$ is the corresponding column mean. The correction is computed from unlabelled scores for the complete query and candidate sets.

The Hungarian method (**?**) finds a permutation π maximising $\sum_i C_{i,\pi(i)}$. For each subject, the final test problem contains 102 queries and 102 candidates, with one recording per candidate. The assigned candidate is placed first in each output row; the remaining candidates follow in stable descending CSLS-score order. Thus, uniqueness is enforced for the first choices across queries, while every row remains a complete candidate ranking. This is transductive inference under a known one-to-one constraint.

For a complete square assignment,

$$\sum_i C_{i,\pi(i)} = 2\sum_i S_{i,\pi(i)} - \sum_i r_i^{(k)} - \sum_v c_v^{(k)}. \tag{5}$$

The last two terms are independent of π. Consequently, CSLS preserves the set of optimal assignments on the same underlying score matrix; it cannot provide a separate assignment-based Top-1 gain in exact arithmetic. Numerical ties can affect the selected representative. The column correction can change the ordering of the remaining candidates and therefore Top-5 and MRR.

We compared $k \in \{1, 5, 10, 20\}$ on training-video validation. The declared selection order was assigned Top-1, then Top-5, then MRR, with numerical rounding for tie resolution. Assigned Top-1 was equal across the tested values, and $k = 1$ had the highest Top-5. For the final mix, assignment reduced validation Top-1 from 29.88% to 29.15%, although CSLS followed by assignment improved Top-5 and MRR (Table 5). Its DEV benefit therefore does not establish a benefit on blind data.

5.2 Validation design and final results

The final comparison split reserves 150 training-video identities using `default_rng(17)`, leaving 850 identities for fitting. The available held-out query and candidate counts are 147, 148, 149, 143, 144 and 124 for subjects 01–06, respectively. Validation metrics are calculated per subject

Evaluation and ranking procedure	Top-1	Top-5	MRR
Training-video validation, independent	29.88%	54.11%	0.417263
Training-video validation, CSLS and assignment	29.15%	58.25%	0.425703
DEV, v7 independent	64.46%	87.01%	0.748086
DEV, v7 CSLS and assignment	72.55%	93.87%	0.819577

Table 5: Ranking procedures applied to the same final group composition within each evaluation protocol. Validation reports subject-macro metrics over six subjects with 124–149 candidates per subject. DEV pools 408 queries from subjects 01–04, with 102 candidates per subject. Absolute validation and DEV scores are not directly comparable. DEV values are recorded in the release; validation values can be recalculated from the packaged holdout scores.

Stage	Recorded configuration	DEV Top-1	DEV Top-5
Static route	Early EEG-to-embedding model	2.70%	—
TRF	Temporal features, Whisper and whitening	21.32%	48.77%
v1	Temporal dual encoder, 16 seeds	34.56%	66.42%
v2	Multi-positive recipe, 24 seeds	54.66%	82.60%
v4	Splicing and three families, twelve seeds each	58.58%	85.78%
v5	Five internal families, twelve seeds each	58.82%	86.27%
v6	Internal/audio fusion, independent ranking	61.27%	86.03%
v7	Internal/audio/visual fusion, CSLS and assignment	72.55%	93.87%

Table 6: Selected development milestones. Values preceding v7 are historically reported and are not independently reproduced from the final release. The rows differ in training recipe, ensemble size and ranking procedure and do not form a controlled single-factor ablation. The historical independent-ranking v6 result must be distinguished from the assigned v6 comparison reported in the text.

and averaged with equal subject weights. Internal validation ensembles use six seeds per family, compared with twelve in the final refit; the supplied audio and visual families use four seeds each. Candidate counts, training-set size and ensemble size therefore differ from DEV. The holdout was repeatedly reused for selection, and DEV was also repeatedly inspected during development.

Final v7 DEV performance is 296/408 Top-1 hits (72.55%), 383/408 Top-5 hits (93.87%) and MRR 0.819577. Subject-specific Top-1 counts are 81, 69, 75 and 71 out of 102. There are 112 Top-1 errors and 25 queries whose true candidate is outside the first five ranks. The available records do not establish content-specific explanations for these errors.

The final mix before CSLS and assignment has 263 Top-1 and 355 Top-5 hits on DEV. The final ranking procedure therefore adds 33 Top-1 hits and 28 Top-5 hits on this set. Under the same CSLS-and-assignment procedure, the recorded v6 internal/audio mix has 293 Top-1 hits (amazing 71.81%), compared with 296 for v7. This comparison separates the change in group composition from changing the ranking procedure; it does not establish a causal or out-of-sample gain from visual fusion.

The historical development record also reports 74.75% DEV Top-1 for an additional DEV-selected v8 family. A subsequent comparison reportedly did not establish an assigned Top-1 gain, although Top-5 and MRR improved. The locked configuration excludes v8. Its checkpoints, score

matrices and comparison details are absent from the release, so neither its reported gain nor a hypothetical effect on blind performance can be independently established here.

6. Signal diagnostics and limits of performance extrapolation

6.1 Cross-subject templates and extrapolated performance

A historical diagnostic constructed stimulus templates from labelled development responses of other subjects. Averaging responses from three other subjects was reported to achieve 71.32% Top-1. This protocol uses candidate-specific EEG responses and their stimulus correspondences, whereas the submitted model learns from training recordings and scores unlabelled test queries. The template result is therefore a separate diagnostic of shared stimulus-related structure, not an estimate of the submitted model's blind accuracy or a bound on its performance.

A linear extrapolation in $1/k$, where k denotes the number of template subjects, reportedly produced an intercept of 84.0% Top-1 as k tends to infinity. The corresponding Top-5 extrapolation was 101.4%, outside the admissible range. This demonstrates that the extrapolation does not provide a calibrated accuracy limit. The fitted relation is not an established EEG noise ceiling, and exceeding the measured template accuracy with v7 is not contradictory: the models, information sources and assignment constraints differ.

The historical template scores, fitted coefficients and uncertainty estimates are not included in the release. Consequently, the reported values are retained as exploratory diagnostics. Establishing a meaningful ceiling would require an explicit measurement model, appropriate repeated observations and uncertainty assessment.

6.2 Learning curves

A historical six-seed experiment varied the fraction of non-holdout training-video identities. Fractions of 25%, 50%, 75% and 100% were reported to yield DEV Top-1 accuracies of 32.35%, 39.95%, 45.34% and 51.72%, respectively. This experiment is distinct from the final 60-member internal ensemble. Its subset assignments, seed-level predictions and uncertainty estimates are absent from the supplied release.

The reported trend is consistent with improved performance when more training data are used, but does not isolate the effect of data quantity. With fixed epochs, larger datasets receive more optimiser updates, and different seeds use different sampled subsets. The four observations do not establish a plateau or an asymptotic accuracy. A controlled follow-up should use shared nested subsets, comparable optimisation budgets and evaluation data excluded from model selection, and should report variation across seeds and subsets.

Interpretation. Template retrieval, extrapolated limits and learning curves address different questions. Their assumptions and evaluation protocols must be specified before they can support claims about available signal or generalisation limits.

7. Negative results and implications for model selection

Recorded experiments identify several extensions that did not improve the corresponding baseline within the tested configurations. Adding bandpower-envelope branches reduced holdout ensemble MRR from 0.345861 to 0.297003, 0.267129 and 0.277581 for the three recorded band config-

urations, decreases of approximately 0.049–0.079 in absolute MRR. These results concern the implemented branches and training recipe; they do not establish that bandpower is generally uninformative for EEG retrieval.

Alternative speech representations and larger encoders produced mixed results. In the recorded wav2vec2 comparison, MRR was 0.336591 against the corresponding 0.345861 reference. AST-based families did not exceed the stored base family's standalone MRR. HuBERT nevertheless added approximately 0.0008 MRR in the extended greedy fusion search. Its exclusion was an engineering trade-off involving an additional dependency, rather than evidence of a uniformly negative contribution. Wider and deeper variants must likewise be interpreted individually; the search does not establish that model capacity is generally harmful.

The stored single-cut splice family has MRR 0.364315, compared with 0.364111 for the two-cut variant. Two splice sets also failed to improve the corresponding recorded configuration. This small two-cut difference does not establish equivalence or a statistically significant disadvantage. Historical trials with finer visual grids, larger PCA representations and random loss windows were reported not to provide a consistent practical gain; their complete implementations and seed-level comparisons are not available in the final release.

The historical CLIP-enriched v3 model was reported to have slightly higher holdout MRR than v2 but DEV Top-1 of 54.41% versus 54.66%. The difference corresponds to one net query among 408, and the historical DEV score matrices are not packaged. In contrast, the released family searches provide numerical evidence for retaining a CLIP-enriched component within a diverse ensemble. Standalone performance and incremental ensemble contribution are therefore distinct selection criteria.

The development process prioritised temporal representations, stimulus-based positive groups, matched augmentation and complementary score families. These choices describe the selected empirical recipe; the sequential experiments do not rank their isolated causal importance. The final inference procedure uses no label-dependent test templates, but repeated DEV inspection and holdout reuse still limit claims about generalisation. Further model selection should use fresh evaluation data and controlled comparisons rather than treating the best reused validation score as an unbiased estimate.

8. Reproducibility and external data

The final release contains the explicit configuration, 60 internal and eight supplied model checkpoints, fitted transforms, reference score matrices, ranking arrays, numerical selection records and SHA-256 checksums. The supplied visual family's four public-data pretraining checkpoints are retained separately. Output arrays are integer 102×102 matrices: each row lists every candidate ID 1001–1102 once, in rank order, and rows preserve the original EEG recording order. The submission archive contains the arrays for subjects 05 and 06. Raw challenge EEG, stimulus videos and extracted frames are excluded; numerical features required by the supplied models are included.

Execution modes and verification. Three execution modes must be distinguished. Frozen-score replay reconstructs the locked rankings and submission archive exactly. Saved-weight inference applies the released models to challenge EEG and can introduce small numerical ranking differences. Retraining refits models and is not expected to reproduce the original weights bit for bit. The release provides all three modes, together with an audit that recomputes family searches and fusion-weight comparisons from held-out training-video scores.

The current release verification record reports 42 passing tests, including default notebook execution, replacing the earlier 30-test snapshot. Its recorded model environment is Python 3.11.9, NumPy 2.2.6 and PyTorch 2.14.1 with CUDA 13.0. Full saved-weight inference on all six subjects, reusing existing stimulus features, agreed with all 612 locked Top-1 assignments. Fused score correlations exceeded 0.99999994. Some lower ranks differed, including one DEV query's Top-5 set; the Top-5 sets for both blind subjects agreed. Agreement with locked predictions measures computational reproduction, not accuracy against withheld labels.

The record additionally documents fresh decoding and public-encoder execution for one real stimulus and a bounded training check using actual challenge EEG across all 68 model members. That check completed 209 optimiser updates, saved and reloaded all members, and reproduced the newly generated fused scores within absolute and relative tolerances of 2×10^{-5}. Most training loops were limited to two updates; the first audio and visual seeds also completed shortened training with inner validation and restoration of the best state. The resulting partial models were verification artefacts and were excluded from the submission.

These checks do not reproduce full-length training to convergence, complete feature extraction for all 1,102 videos, public-data pretraining or a first download with an empty model cache. The execution record and its limits are retained in `validation/execution_smoke.json` and `validation/weight_reproduction.json`. The verification claims refer to that recorded environment and exercised paths, not to every hardware or dependency configuration.

External models and data. The internal public encoders are Whisper-base (2), DINOv2-base (3) and CLIP ViT-B/32 (1); their model identifiers and exact revisions are pinned in the release. The supplied audio model additionally uses Whisper-small features and a PCA transform fitted on public SparrKULee/ICASSP 2023 Auditory EEG Decoding Challenge speech windows (4). Its trainable dual encoder uses the selected scratch configuration. The supplied visual model is fine-tuned from checkpoints declared to have been pretrained on THINGS-EEG2 (5).

The release declares CC BY-NC 4.0 for the supplied speech-data source and CC BY 4.0 for THINGS-EEG2. These are source-specific declarations, not a blanket licence for every model, transform or image asset in the pipeline. Supplier code, manifests, transforms and checkpoints preserve the recorded provenance; exclusive dataset use and the complete original public-data pretraining lineage were not independently audited. These sources are distinct from the historical EMD proxy discussed earlier.

Reproducibility limits. Exact replay establishes the ability to regenerate the submitted output from frozen scores. It does not independently establish historical training results, the complete provenance of external weights or blind generalisation. Historical test-suite counts from earlier pipeline versions are therefore not used as evidence for the final release.

9. Conclusion

The final system combines temporal audiovisual representations, subject-aware EEG encoding, multi-positive contrastive training, matched temporal splicing and complementary score families. On the repeatedly inspected DEV set, the released v7 ranking procedure achieves 296/408 Top-1 hits (72.55%), 383/408 Top-5 hits (93.87%) and MRR 0.819577. These results concern previously unseen stimuli for known subjects and include transductive one-to-one assignment.

The study documents an empirical development process, including discarded information paths, negative results and the numerical evidence behind the selected ensemble. The sequential experiments do not isolate the causal contribution of each design choice. Reused validation data, repeated DEV inspection and incomplete historical experiment artefacts limit the strength of generalisation claims. Performance on the withheld subject-05/06 correspondences remains unknown in the supplied release.

Further evaluation should use data excluded from model selection, controlled training-budget comparisons and an audited splice-label implementation. These checks are needed to assess whether the observed development gains persist beyond the evaluated protocols.

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